Tutorial 3 - Digital Systems

1. Simplify the Boolean expressions using 3-variable K-maps.

- (a) $F(x, y, z) = (3, 4, 6, 7)$
- (b) $F(x, y, z) = (0, 2, 4, 5, 6, 7)$
- (c) $F(x, y, z) = (1, 3, 5, 7, 6)$

2. Simplify the following Boolean function using a Karnaugh map (K-map) and identify the prime implicants and essential prime implicants:

$$Y = \sum(0, 2, 3, 4, 9, 10, 12, 13) + d(6, 8, 14)$$

3. A logic circuit realizes the function F(a,b,c,d) = a'b' + a'cd + ac'd + ab'd'. Assuming that a = c never occurs when b = d = 1, find a simplified expression for F.

4. Consider the Boolean function $z(a, b, c)$ in Figure 1. Which of the following minterm expansions represents this circuit ?

(a) $z = \Sigma(0, 1, 3, 7)$

(b) $z = \Sigma(1, 4, 5, 6, 7)$

(c) $z = \Sigma(2, 4, 5, 6, 7)$

(d) $z = \Sigma(2, 3, 5)$

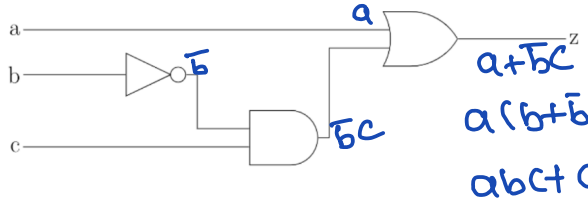


Figure 1: Logic Circuit

5. A plot of minterms for a four variable function is given in Figure 2. Draw the K-map and simplify.

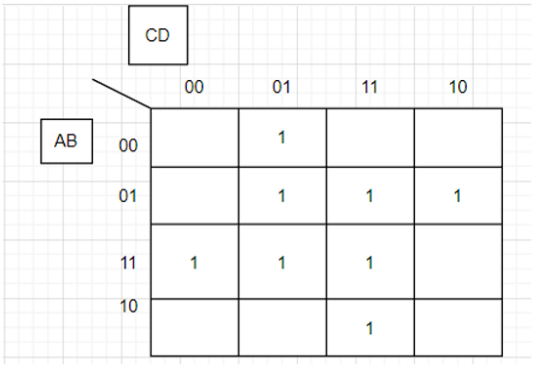


Figure 2: Plot of minterms of a function of four variables

6. A logic circuit realizing a function f has four inputs *a, b, c* and *d*. The three inputs *a, b*, and *c* are the binary representations of the digits 0 through 7 with *a* being the most significant bit. The input *d* is an odd-parity bit; that is, the value of *d* is such that *a, b, c,* and *d* always contain an odd number of 1s. (For example, the digit 1 is represented by abc = 001 and d = 0, and the digit 3 is represented by abcd = 0111.) The function f has value 1 if the input digit is a prime number. Cells which correspond to an even number of 1s (including 0000) represent don't cares. Draw a Karnaugh map for f and simplify to get the minimum sum of products for f.

7. The plot of minterms for a four-variable Boolean function is given in Figure 3. Find the minimal POS form using a K-map of the function.



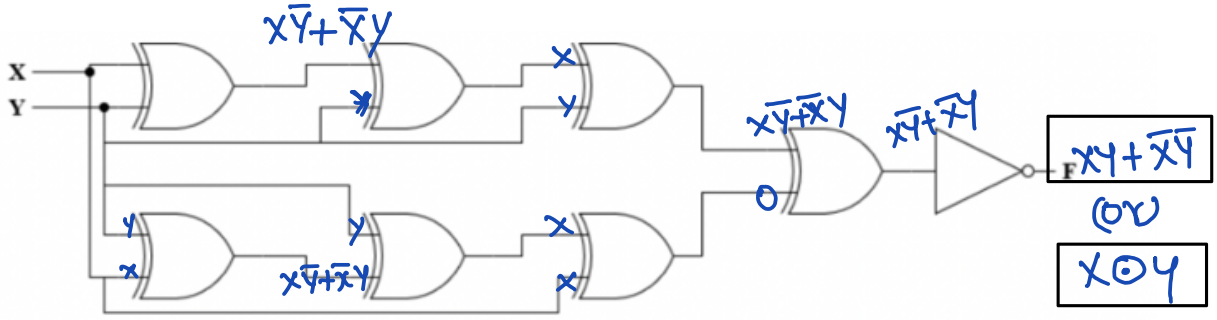
Figure 3: Plot of minterms for a four-variable function

8. For $F(a, b, c, d) = \pi(0, 1, 3, 7)$, the number of terms in the minimal SOP form is given by 'X'. Find the value of X. Explain your solution.

9. Draw a logic diagram using only two-input NOR gates to implement the following:

$$F(A, B, C, D) = (A \oplus B)'(C \oplus D)$$

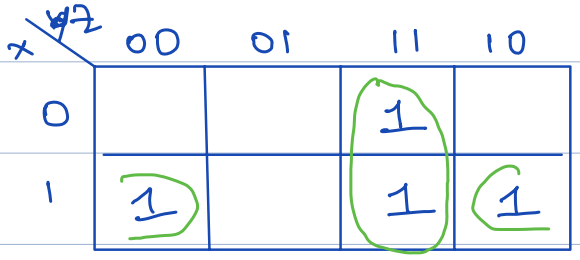
10. What is the output of the circuit shown below?



1. Simplify the Boolean expressions using 3-variable K-maps.

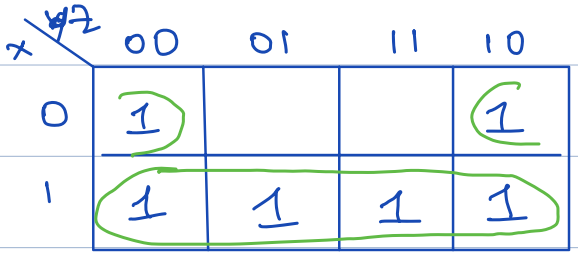
- (a) $F(x, y, z) = (3, 4, 6, 7)$
- (b) $F(x, y, z) = (0, 2, 4, 5, 6, 7)$
- (c) $F(x, y, z) = (1, 3, 5, 7, 6)$

(a)



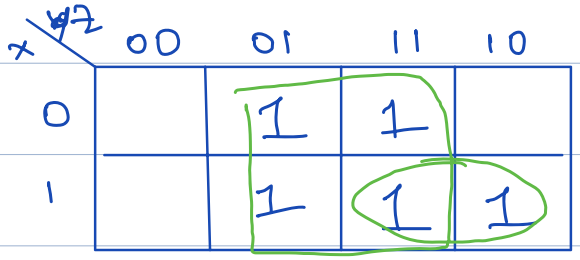
$= yz + x\bar{z}$

(b)



$= x + z'$

(c)



$= z + xy$

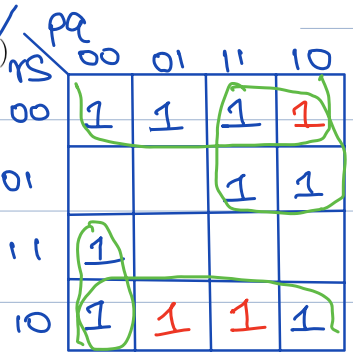
2. Simplify the following Boolean function using a Karnaugh map (K-map) and identify the prime implicants and essential prime implicants:

$$Y = \sum(0, 2, 3, 4, 9, 10, 12, 13) + d(6, 8, 14)$$

$Y = s' + pr' + p'q'r$

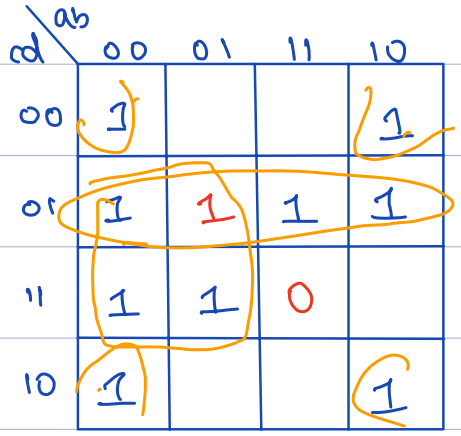
(prime implicants: $p'q'r$, pr', s')

(essential: $p'q'r$, pr', s')



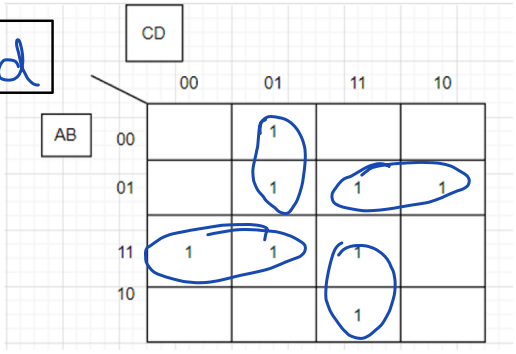
3. A logic circuit realizes the function F(a,b,c,d) = a'b' + a'cd + ac'd + ab'd'. Assuming that a = c never occurs when b = d = 1, find a simplified expression for F.

$F = c'd + a'd + b'd'$



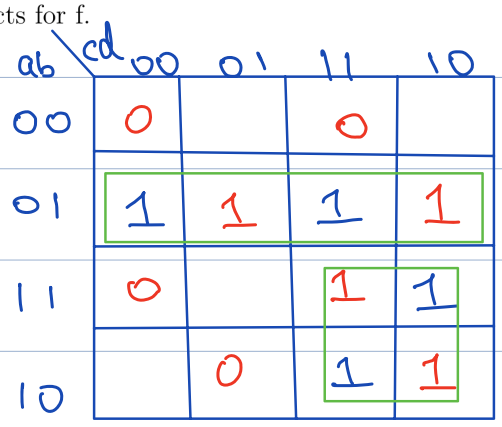
5. A plot of minterms for a four variable function is given in Figure 2. Draw the K-map and simplify.

$abc' + acd + a'bc + a'c'd$



6. A logic circuit realizing a function f has four inputs *a, b, c* and *d*. The three inputs *a, b*, and *c* are the binary representations of the digits 0 through 7 with *a* being the most significant bit. The input *d* is an odd-parity bit; that is, the value of *d* is such that *a, b, c,* and *d* always contain an odd number of 1s. (For example, the digit 1 is represented by abc = 001 and d = 0, and the digit 3 is represented by abcd = 0111.) The function f has value 1 if the input digit is a prime number. Cells which correspond to an even number of 1s (including 0000) represent don't cares. Draw a Karnaugh map for f and simplify to get the minimum sum of products for f.

$a'b + ac$



7. The plot of minterms for a four-variable Boolean function is given in Figure 3. Find the minimal POS form using a K-map of the function.

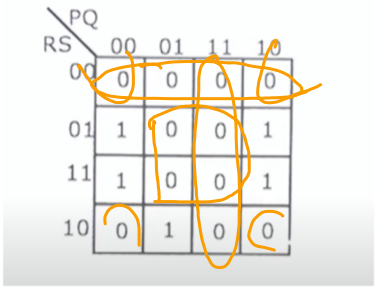


Figure 3: Plot of minterms for a four-variable function

$F' = (r's' + qs + pq + q's')$

$F = (r+s)(q+s)(p'+q)(q'+s')$

8. For $F(a, b, c, d) = \pi(0, 1, 3, 7)$, the number of terms in the minimal SOP form is given by 'X'. Find the value of X. Explain your solution.

$F = \Sigma(2,4,5,6,8,9,10,11,12,13,14,15)$

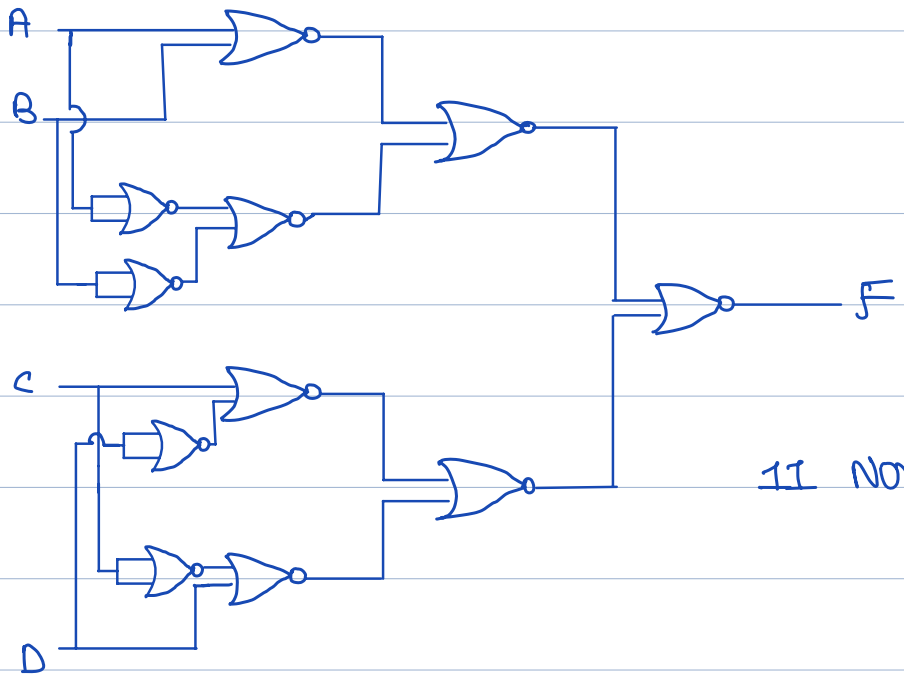
$F = a + bc' + cd'$

$X = 3$

ab \ cd	00	01	11	10
00				1
01	1	1		1
11	1	1	1	1
10	1	1	1	1

9. Draw a logic diagram using only two-input NOR gates to implement the following:

$$F(A, B, C, D) = (A \oplus B)'(C \oplus D)$$



11 NOR gates